

Equality in the Positive Square-Energy Bound

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Abstract

Let $s^+(G)$ be the sum of the squares of the positive adjacency eigenvalues of a graph G . Liu, Tang, and Zhang recently proved the Elphick–Farber–Goldberg–Wocjan lower bound $s^+(G) \geq |V(G)| - 1$ for connected graphs. We determine its equality case: a connected graph G of order n satisfies $s^+(G) = n - 1$ if and only if G is a tree. We first trace equality through the doubly nonnegative inequality of Liu–Tang–Zhang. Their folding and cut-vertex induction, supplemented by their averaged vertex-deletion estimate, forces every block of an equality graph to be complete. We then prove that the positive spectral part of a tree is strictly positive between vertices having a common neighbor. A spanning-tree variational argument excludes every non-tree block graph.

1 Introduction

All graphs are finite, simple, and undirected. Let $A = A(G)$ have eigenvalues $\lambda_1, \dots, \lambda_n$, and define

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2.$$

Writing the positive and negative spectral parts of A as

$$A = P - B, \quad P = A_+ \succeq 0, \quad B = A_- \succeq 0, \quad PB = 0,$$

we have $s^+(G) = \text{tr } P^2$, $s^-(G) = \text{tr } B^2$, and

$$s^+(G) + s^-(G) = \text{tr } A^2 = 2|E(G)|. \tag{1}$$

Elphick, Farber, Goldberg, and Wocjan conjectured that $\min\{s^+(G), s^-(G)\} \geq n - 1$ for every connected graph [3]. Liu, Tang, and Zhang recently proved this theorem by means of a sharp inequality over the doubly nonnegative cone [5]. Akbari, Kumar, Mohar, Pragada, and Zhang asked for the equality cases and, in particular, conjectured that equality for s^+ characterizes trees [1, Conjecture 9.2(i)]. We prove that conjecture.

Theorem 1. *Let G be a connected graph on n vertices. Then*

$$s^+(G) = n - 1 \iff G \text{ is a tree.}$$

The equality trace in Section 3 uses the ingredients of the proof of the Liu–Tang–Zhang doubly nonnegative inequality, together with an equality comparison not needed for their theorem. We include the details, both to fix the direction of the inequality and to make clear how the complete-block conclusion follows.

2 The doubly nonnegative inequality

A real symmetric matrix M is *doubly nonnegative* if $M \succeq 0$ and M is entrywise nonnegative. For a connected graph H , put

$$q(H) = 2|E(H)| - |V(H)| + 1, \quad S(H, M) = \sum_{uv \in E(H)} \sqrt{M_{uv}}, \quad T(M) = \mathbf{1}^T M \mathbf{1}.$$

The orientation of the central inequality is important.

Theorem 2 (Liu–Tang–Zhang [5]). *If H is connected and M is a doubly nonnegative matrix indexed by $V(H)$, then*

$$4S(H, M)^2 \leq q(H)T(M). \quad (2)$$

We recall the parts of its proof used below. First, every nonedge entry can be folded onto the diagonal:

$$\mathcal{F}_H(M) = M + \sum_{\substack{\{u,v\} \notin E(H) \\ u \neq v}} M_{uv}(e_u - e_v)(e_u - e_v)^T. \quad (3)$$

The sum takes each unordered nonedge once. The folded matrix is doubly nonnegative, vanishes on nonedges, has the same edge entries as M , and has the same value of T , since $(e_u - e_v)^T \mathbf{1} = 0$.

Suppose next that an edge-supported pair (H, M) is split at a cut vertex into connected induced graphs H_1, H_2 . A Gram representation of M and orthogonal projection of the cut-vertex vector produce doubly nonnegative matrices M_1, M_2 such that

$$q(H) = q(H_1) + q(H_2), \quad S(H, M) = S(H_1, M_1) + S(H_2, M_2), \quad T(M) = T(M_1) + T(M_2). \quad (4)$$

The induction bounds for the two pairs are combined by scalar Cauchy–Schwarz.

Finally, suppose H has $h \geq 3$ vertices and no cut vertex. Then $H - v$ is connected for every v . For an edge-supported M , write

$$d_0 = \text{tr } M, \quad w = \sum_{uv \in E(H)} M_{uv}, \quad T = d_0 + 2w, \quad q = q(H).$$

Averaging the induction hypothesis over the h vertex deletions gives

$$4S(H, M)^2 \leq (q - 1)T + \frac{q - 1}{h - 2}d_0. \quad (5)$$

Independently, Cauchy–Schwarz on the $\ell = |E(H)|$ edges gives the flat estimate

$$4S(H, M)^2 \leq 4\ell w. \quad (6)$$

The proof of Theorem 2 divides according as

$$2(h - 1)w \leq qd_0 \quad (7)$$

or the reverse strict inequality holds. In the first case (6) and $4\ell = 2q + 2(h - 1)$ prove (2). In the second case, put $\beta = \ell - h + 1$. Then $q - 1 = (h - 2) + 2\beta$, and simplicity gives

$$q(h - 2) - 2\beta(h - 1) = (h - 1)(h - 2) - 2\beta \geq 0.$$

Multiplying the strict case hypothesis $qd_0 < 2(h - 1)w$ by $h - 2$ therefore gives

$$2\beta(h - 1)d_0 \leq q(h - 2)d_0 < 2(h - 1)(h - 2)w,$$

and hence $\beta d_0 < (h - 2)w$, or equivalently $(q - 1)d_0 / (h - 2) < T$. Thus (5) gives the strict inequality $4S^2 < qT$. These are precisely the folding, cut, averaged, and flat ingredients from the proof in [5].

3 Equality through folding and cuts

We now record the equality consequence needed later. A *block* is a maximal subgraph with no cut vertex; thus a bridge is a block isomorphic to K_2 , while every block of order at least three is 2-connected.

Proposition 3 (Equality trace). *Let H be a connected graph with at least one edge, let M be doubly nonnegative, and suppose*

$$4S(H, M)^2 = q(H)T(M) > 0. \quad (8)$$

Set $N = \mathcal{F}_H(M)$ and $\lambda = T(M)/q(H)$. Then every block C of H is either a bridge or a complete graph, and

$$N = \lambda \sum_C \frac{q(C)}{|V(C)|^2} J_{V(C)}, \quad (9)$$

where each all-ones matrix is embedded on $V(C)$. In particular, for every edge uv in a block C ,

$$\sqrt{M_{uv}} = \sqrt{\lambda} \frac{q(C)}{2|E(C)|}. \quad (10)$$

Proof. Folding preserves S and T , so (H, N) is again tight and is supported on the diagonal and edges. We trace the induction proving Theorem 2.

At a cut vertex, use (4) and abbreviate $q_i = q(H_i)$ and $T_i = T(N_i)$. The induction and scalar Cauchy–Schwarz give

$$2S(H, N) \leq \sqrt{q_1 T_1} + \sqrt{q_2 T_2} \leq \sqrt{(q_1 + q_2)(T_1 + T_2)}.$$

Equality at the two ends forces equality in both descendant induction bounds and equality in scalar Cauchy–Schwarz. Hence

$$\frac{T_1}{q_1} = \frac{T_2}{q_2} = \frac{T(N)}{q(H)} = \lambda. \quad (11)$$

Repeating the split therefore leaves tight descendant pairs with the same normalization $T_C = \lambda q(C)$. The terminal descendants are bridges or graphs of order at least three with no cut vertex.

We will also use the entrywise content of the cut construction. Let z be the cut vertex, let U and W be the two sides away from z , and represent N by Gram vectors g_x . Since N is edge-supported, the spans

$$\mathcal{U} = \text{span}\{g_u : u \in U\}, \quad \mathcal{W} = \text{span}\{g_w : w \in W\}$$

are orthogonal. Decompose

$$g_z = p_U + p_W + p_0, \quad p_U \in \mathcal{U}, \quad p_W \in \mathcal{W}, \quad p_0 \perp \mathcal{U} + \mathcal{W}.$$

Use $p_U + p_0$ as the cut vector on the U -side and p_W on the W -side. Inner products from z to either side are unchanged, all entries inside a side are unchanged, and

$$\|p_U + p_0\|^2 + \|p_W\|^2 = \|g_z\|^2.$$

Consequently the two descendant matrices, embedded back into $V(H)$, add entrywise to N . In particular, every edge entry is preserved in its unique descendant. Repeating this construction terminates precisely at the blocks of H , independently of the order in which cut vertices are split.

Consider such a no-cut descendant (C, X) , and write $h = |V(C)|$, $\ell = |E(C)|$, $q = q(C)$, $d_0 = \text{tr } X$, and $w = \sum_{uv \in E(C)} X_{uv}$. Equality cannot pass through the strict second case of the

Liu–Tang–Zhang argument recalled after (7). It therefore lies in the first case, and equality in the final bound forces equality both in the flat edge Cauchy–Schwarz estimate and at the threshold:

$$X_{uv} = r \quad (uv \in E(C)), \quad 2(h-1)w = qd_0. \quad (12)$$

Here $r > 0$: the tight terminal descendant has $T = \lambda q > 0$, and hence $S > 0$. Since $w = \ell r$, equality in (6) is explicit.

The averaged estimate (5) remains valid independently of which case proves the theorem. Comparing it with $4S^2 = qT$ gives

$$d_0 \geq \frac{h-2}{q-1}T. \quad (13)$$

On the other hand, the threshold equality in (12), together with $T = d_0 + 2w$, gives

$$d_0 = \frac{h-1}{q+h-1}T. \quad (14)$$

Comparing (13) and (14) and cross-multiplying yields

$$(h-1)(q-1) \geq (h-2)(q+h-1), \quad \text{hence} \quad q \geq (h-1)^2.$$

Simplicity supplies the reverse bound

$$q = 2\ell - h + 1 \leq h(h-1) - h + 1 = (h-1)^2.$$

Thus $\ell = \binom{h}{2}$ and $C = K_h$. This is the crucial complete-block conclusion; it comes from retaining the averaged estimate even though the flat estimate is the one used in the first case.

It remains to identify the terminal matrix rather than only its edge entries. For $C = K_h$, we have $q = (h-1)^2$, and equality in the flat estimate gives

$$qT = 4\binom{h}{2}^2 r, \quad T = h^2 r.$$

Equation (14) gives $d_0 = hr$. Write the diagonal entries of X as $rb_1, \dots, rb_h$. Every 2×2 principal minor is nonnegative, so $b_i b_j \geq 1$ for $i \neq j$, while $\sum_i b_i = h$. These conditions force $b_i = 1$ for every i : if some $b_i < 1$, every other $b_j \geq b_i^{-1}$ and the sum exceeds h ; if no $b_i < 1$, the sum condition is decisive. Consequently $X = rJ_h$. Since $T = \lambda q$, we obtain

$$X = \lambda \frac{(h-1)^2}{h^2} J_h = \lambda \frac{q(C)}{|V(C)|^2} J_{V(C)}. \quad (15)$$

For a bridge $C = K_2$, the base case is the same conclusion. Indeed, if $X = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ is tight, then

$$4c = a + b + 2c,$$

whereas positive semidefiniteness gives $2c \leq 2\sqrt{ab} \leq a + b$. Equality follows throughout, so $a = b = c$, and $X = (\lambda/4)J_2$ because $q(K_2) = 1$ and $T = \lambda$.

The entrywise cut identity proved above recombines the terminal matrices by adding their embedded block contributions; the only overlap is on cut-vertex diagonal entries, where the contributions add to the parent diagonal. Induction on the number of cut splits now gives (9). Finally, an edge belongs to a unique block. For a bridge, $\sqrt{q(C)}/|V(C)| = 1/2 = q(C)/(2|E(C)|)$; for $C = K_h$, both quantities equal $(h-1)/h$. Taking the square root of its coefficient in (9) proves (10). $\square$

We apply the proposition to the negative spectral part. This also explains why signs and absolute values cannot be omitted from the initial reduction.

Corollary 4. *Suppose G is connected, has $n \geq 2$ vertices and m edges, and satisfies $s^+(G) = n - 1$. Then every block of G is complete. More precisely, if an edge uv lies in a block C , then*

$$B_{uv} = -\frac{q(C)}{2|E(C)|}. \quad (16)$$

Proof. Put $q = 2m - n + 1$. By (1), $s^-(G) = q$. The Schur product theorem [4, Theorem 7.5.3] makes $M = B \circ B$ doubly nonnegative. Moreover

$$\text{tr}(AB) = \text{tr}((P - B)B) = -\text{tr } B^2 = -s^-(G),$$

and A has two symmetric unit entries for each edge. Therefore

$$\frac{s^-(G)}{2} = -\sum_{uv \in E(G)} B_{uv} \leq \sum_{uv \in E(G)} |B_{uv}|.$$

Theorem 2, applied to $B \circ B$, now gives the requested starting chain

$$s^-(G)^2 \leq 4 \left(\sum_{uv \in E(G)} |B_{uv}| \right)^2 \leq q \mathbf{1}^T (B \circ B) \mathbf{1} = q s^-(G). \quad (17)$$

The first inequality follows exactly from the displayed trace identity and the triangle inequality on the edge entries. Since $s^-(G) = q > 0$, both inequalities in (17) are equalities. Equality in the triangle inequality gives

$$B_{uv} \leq 0 \quad (uv \in E(G)), \quad (18)$$

and $B \circ B$ is an equality matrix for Theorem 2. Its normalization in Proposition 3 is $\lambda = T(B \circ B)/q = s^-(G)/q = 1$. The proposition makes every block a bridge or a complete graph and, because folding preserves edge entries, gives $|B_{uv}| = q(C)/(2|E(C)|)$. Equation (18) proves (16). $\square$

Thus a putative equality graph is a *block graph*: its blocks are complete. Notice that this conclusion does not assert, falsely, that a leaf partition turns all cyclic blocks into triangles. Blocks of arbitrary order remain possible at this stage and will all be excluded at once.

4 Strict positivity between siblings

For a symmetric matrix K , let K_+ denote its positive spectral part. The next lemma is the strict input for the final variational argument.

Lemma 5 (Sibling positivity). *Let T be a tree. If distinct vertices u, v have a common neighbor in T , then*

$$(A(T)_+)_{uv} > 0.$$

Proof. Choose a bipartition $L \sqcup R$ of T with $u, v \in L$, and write

$$A(T) = \begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix}, \quad Q = XX^T. \quad (19)$$

For $r \in R$, let $j_r \in \mathbb{R}^L$ be the indicator of $N_T(r)$. For every $a > 0$,

$$Q + aI = aI + \sum_{r \in R} j_r j_r^T. \quad (20)$$

We prove

$$(Q + aI)_{uv}^{-1} < 0. \quad (21)$$

Regard T itself as the incidence graph between the variable vertices L and the neighborhood hyperedges $N_T(r)$, $r \in R$. It is a tree. Root it at a common neighbor r_0 of u and v , and eliminate branches in postorder. We spell out the Schur calculation, including singleton neighborhoods.

Consider a nonroot neighborhood node r whose parent variable is $p \in L$. After all branches below it have been eliminated, its child variables $C = N_T(r) \setminus \{p\}$ carry a positive diagonal matrix D_C : positivity starts with the term aI , and the calculation below shows that every eliminated child neighborhood adds a positive number to its articulation diagonal. On $\{p\} \cup C$, the contribution of r together with those effective child diagonals has the form $J + D_C$ on the child coordinates, with the p diagonal of J kept at this step. Eliminating C changes the p diagonal by

$$\begin{aligned} 1 - \mathbf{1}_C^T (D_C + J_C)^{-1} \mathbf{1}_C &= 1 - \frac{\mathbf{1}_C^T D_C^{-1} \mathbf{1}_C}{1 + \mathbf{1}_C^T D_C^{-1} \mathbf{1}_C} \\ &= \frac{1}{1 + \mathbf{1}_C^T D_C^{-1} \mathbf{1}_C} > 0, \end{aligned} \quad (22)$$

where Sherman–Morrison was used. If $C = \emptyset$, so that r is a singleton leaf neighborhood, the value in (22) is understood as 1; thus such leaves are retained rather than silently discarded. This proves inductively both the asserted diagonal form and its strict positivity.

After every branch below r_0 is eliminated, the Schur complement on the root neighborhood $N_T(r_0)$ is

$$D_0 + J, \quad D_0 = \text{diag}(d_x : x \in N_T(r_0)), \quad d_x > 0.$$

Schur complementation preserves the corresponding entries of the inverse. Another application of Sherman–Morrison gives, for distinct $x, y \in N_T(r_0)$,

$$(D_0 + J)_{xy}^{-1} = -\frac{d_x^{-1} d_y^{-1}}{1 + \sum_{z \in N_T(r_0)} d_z^{-1}} < 0. \quad (23)$$

Taking $x = u$ and $y = v$ proves (21). This argument is valid without any rank assumption on X because $a > 0$ makes every matrix being eliminated positive definite.

The scalar identity

$$\sqrt{t} = \frac{1}{\pi} \int_0^\infty \frac{t}{t+a} a^{-1/2} da \quad (t \geq 0)$$

and spectral functional calculus [2, Chapter V] give, for $u \neq v$,

$$(Q^{1/2})_{uv} = -\frac{1}{\pi} \int_0^\infty a^{1/2} (Q + aI)_{uv}^{-1} da. \quad (24)$$

The integral converges entrywise. As $a \downarrow 0$, $(Q + aI)^{-1} = a^{-1} \Pi_{\ker Q} + O(1)$, so after multiplication by $a^{1/2}$ the possible singularity is $O(a^{-1/2})$. As $a \rightarrow \infty$, an off-diagonal resolvent entry is $O(a^{-2})$, and the integrand is $O(a^{-3/2})$. By (21), the integrand after the minus sign is strictly positive for every $a > 0$. Hence $(Q^{1/2})_{uv} > 0$.

Finally, standard functional calculus applied to (19) gives

$$|A(T)| = \begin{pmatrix} Q^{1/2} & 0 \\ 0 & (X^T X)^{1/2} \end{pmatrix}, \quad A(T)_+ = \frac{1}{2} \begin{pmatrix} Q^{1/2} & X \\ X^T & (X^T X)^{1/2} \end{pmatrix}.$$

Thus $(A(T)_+)_{uv} = \frac{1}{2}(Q^{1/2})_{uv} > 0$, as claimed. $\square$

5 The variational finish

We use the following standard identity, included for completeness.

Lemma 6. *For every real symmetric matrix K ,*

$$\mathrm{tr}(K_+^2) = \max_{Y \succeq 0} \{2 \mathrm{tr}(KY) - \mathrm{tr}(Y^2)\}. \quad (25)$$

Proof. Since $K = K_+ - K_-$ and $K_-, Y \succeq 0$, $\mathrm{tr}(K_- Y) \geq 0$. Therefore

$$2 \mathrm{tr}(KY) - \mathrm{tr}(Y^2) \leq 2 \mathrm{tr}(K_+ Y) - \mathrm{tr}(Y^2) = \mathrm{tr}(K_+^2) - \|Y - K_+\|_F^2.$$

Equality is attained at $Y = K_+$. $\square$

Proposition 7. *Every connected block graph G on n vertices that is not a tree satisfies*

$$s^+(G) > n - 1.$$

Proof. Consider the augmented incidence graph whose nodes are all vertices and all blocks of G , with a vertex joined to each block containing it. For a block graph this incidence graph is a tree. Root it at an arbitrary graph vertex. In each complete block C , let p_C be its unique vertex nearest the root. Retain only the star edges $p_C x$ for $x \in V(C) \setminus \{p_C\}$. For a bridge this retains its sole edge. The union of all retained stars is connected and has

$$\sum_C (|V(C)| - 1) = n - 1$$

edges, so it is a spanning tree T of G .

Every deleted edge uv belongs to a complete block and joins two distinct vertices of $V(C) \setminus \{p_C\}$. Thus u and v have the common neighbor p_C in T . Write

$$A(G) = A(T) + M, \quad P = A(T)_+,$$

where M is the adjacency matrix of the deleted edges. Testing $Y = P$ in (25) gives

$$\begin{aligned} s^+(G) &\geq 2 \mathrm{tr}((A(T) + M)P) - \mathrm{tr}(P^2) \\ &= s^+(T) + 2 \mathrm{tr}(MP) \\ &= n - 1 + 4 \sum_{uv \in E(G) \setminus E(T)} P_{uv}. \end{aligned} \quad (26)$$

Here $s^+(T) = |E(T)| = n - 1$, because the spectrum of the bipartite graph T is symmetric and $s^+(T) + s^-(T) = 2|E(T)|$. Every summand in (26) is strictly positive by Lemma 5. Since a non-tree block graph has at least one deleted edge, the inequality is strict. $\square$

Proof of Theorem 1. First let $n = 1$. The unique connected graph is K_1 , which is a tree, and $s^+(K_1) = 0 = n - 1$.

Now let $n \geq 2$ and suppose $s^+(G) = n - 1$. Corollary 4 shows that every block of G is complete, so G is a block graph. If it were not a tree, Proposition 7 would give $s^+(G) > n - 1$, a contradiction. Hence G is a tree.

Conversely, if G is a tree, then it is bipartite, so its nonzero adjacency eigenvalues occur in opposite pairs. Therefore

$$s^+(G) = \frac{1}{2} \operatorname{tr} A(G)^2 = |E(G)| = n - 1.$$

This also covers K_1 directly, and completes the proof. $\square$

References

- [1] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [2] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics 169, Springer, New York, 1997.
- [3] C. Elphick, M. Farber, F. Goldberg, and P. Wocjan, *Conjectured bounds for the sum of squares of positive eigenvalues of a graph*, Discrete Math. 339 (2016), 2215–2223.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2013.
- [5] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.